📊 E-Commerce Customer Analysis

📌 Project Overview

This markdown cell introduces the project's focus on e-commerce data analysis

The analysis will examine three key areas:

1. Sales trends - to understand revenue patterns over time

2. Customer behavior - to identify purchasing patterns and preferences

3. Profitability insights - to determine which products/customers drive profit

The ultimate purpose is to provide actionable insights for business decisions

🎯 Business Problem

This section defines the core business challenge being addressed

E-commerce companies collect extensive transaction data

The challenge is extracting meaningful insights from this raw data

Without proper analysis, businesses cannot identify:

- Profitable customer segments

- High-performing products

- Emerging market trends

E-commerce businesses generate large volumes of transactional data, but without analysis it is difficult to identify profitable segments, products, and trends.

In [18]:
```
import pandas as pd
import plotly.express as px # Data visualization library
import plotly.graph_objects as go # Graphical visuals
import plotly.io as pio  # Graph template
import plotly.colors as colors
pio.templates.default="plotly_white"
```

data=pd.read_csv('superstore.csv',encoding= 'latin- 1') data.head()

📂 Dataset Overview

The dataset contains order-level e-commerce data including sales, profit, category, sub-category, customer segment, and order date.

In [5]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10800 entries, 0 to 10799
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Row ID         10800 non-null  object 
 1   Order ID       10800 non-null  object 
 2   Order Date     9994 non-null   object 
 3   Ship Date      9994 non-null   object 
 4   Ship Mode      9994 non-null   object 
 5   Customer ID    9994 non-null   object 
 6   Customer Name  9994 non-null   object 
 7   Segment        9994 non-null   object 
 8   Country        9994 non-null   object 
 9   City           9994 non-null   object 
 10  State          9994 non-null   object 
 11  Postal Code    9983 non-null   float64
 12  Region         9994 non-null   object 
 13  Product ID     9994 non-null   object 
 14  Category       9994 non-null   object 
 15  Sub-Category   9994 non-null   object 
 16  Product Name   9994 non-null   object 
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   float64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(5), object(16)
memory usage: 1.7+ MB
```

In [7]:
```
data['Order Date']=pd.to_datetime(data['Order Date'])
data['Ship Date']=pd.to_datetime(data['Ship Date'])
```

In [8]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10800 entries, 0 to 10799
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype         
---  ------         --------------  -----         
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 1   Order ID       10800 non-null  object        
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   datetime64[ns]
 4   Ship Mode      9994 non-null   object        
 5   Customer ID    9994 non-null   object        
 6   Customer Name  9994 non-null   object        
 7   Segment        9994 non-null   object        
 8   Country        9994 non-null   object        
 9   City           9994 non-null   object        
 10  State          9994 non-null   object        
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 12  Region         9994 non-null   object        
 13  Product ID     9994 non-null   object        
 14  Category       9994 non-null   object        
 15  Sub-Category   9994 non-null   object        
 16  Product Name   9994 non-null   object        
 17  Sales          9994 non-null   float64       
 18  Quantity       9994 non-null   float64       
 19  Discount       9994 non-null   float64       
 20  Profit         9994 non-null   float64       
dtypes: datetime64[ns](2), float64(5), object(14)
memory usage: 1.7+ MB
```

In [10]:
```
data['Order Month']=data['Order Date'].dt.month
data['Order Year']= data['Order Date'].dt.year
data['Order by Week']= data['Order Date'].dt.dayofweek
```

In [11]:
```
data.head()
```

Out[11]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2017-138688	2017-06-12	2017-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 24 columns

In [13]:
```
data['Order Month'] = data['Order Month'].astype('Int64')
data['Order Year'] = data['Order Year'].astype('Int64')
data['Order by Week'] = data['Order by Week'].astype('Int64')
```

In [14]:
```
data.head()
```

Out[14]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2017-138688	2017-06-12	2017-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 24 columns

In [15]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10800 entries, 0 to 10799
Data columns (total 24 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         10800 non-null  object
 1   Order ID       10800 non-null  object
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   datetime64[ns]
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
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 13  Product ID     9994 non-null   object
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 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   float64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
 21  Order Month    9994 non-null   Int64
 22  Order Year     9994 non-null   Int64
 23  Order by Week  9994 non-null   Int64
dtypes: Int64(3), datetime64[ns](2), float64(5), object(14)
memory usage: 2.0+ MB
```

🔍 Exploratory Data Analysis (EDA)

Exploratory analysis is performed to identify patterns and trends in sales, profit, and customer behavior.

In [16]:
```
sales_by_month = data.groupby('Order Month')['Sales'].sum().reset_index()
sales_by_month
```

Out[16]:

	Order Month	Sales
0	1	94924.8356
1	2	59751.2514
2	3	205005.4888
3	4	137762.1286
4	5	155028.8117
5	6	152718.6793
6	7	147238.0970
7	8	159044.0630
8	9	307649.9457
9	10	200322.9847
10	11	352461.0710
11	12	325293.5035

In [20]:
```
fig=px.line(sales_by_month,
            x='Order Month',
            y='Sales',
            title='Monthly Sales Analysis')
fig.show()
```

In [21]:
```
data.head()
```

Out[21]:

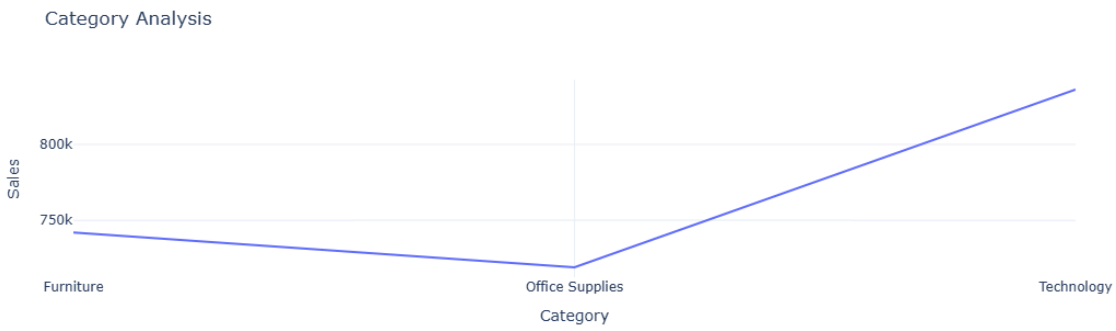
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
1	2	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Her
2	3	CA-2017-138688	2017-06-12	2017-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	
3	4	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau
4	5	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Lau

5 rows × 24 columns

📌 Sales by Category

This analysis compares total sales across different product categories.

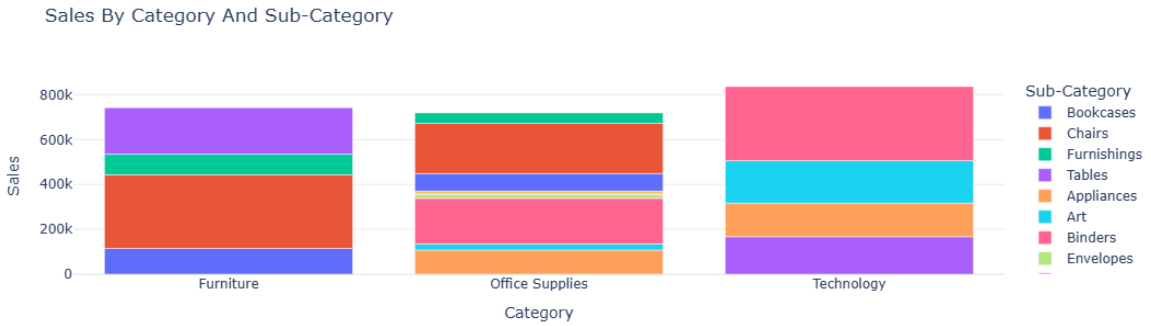
In [22]:
```
sales_by_category= data.groupby('Category')['Sales'].sum().reset_index()
fig=px.line(sales_by_category,
            x= 'Category',
            y= 'Sales',
            title= 'Category Analysis')
fig.show()
```



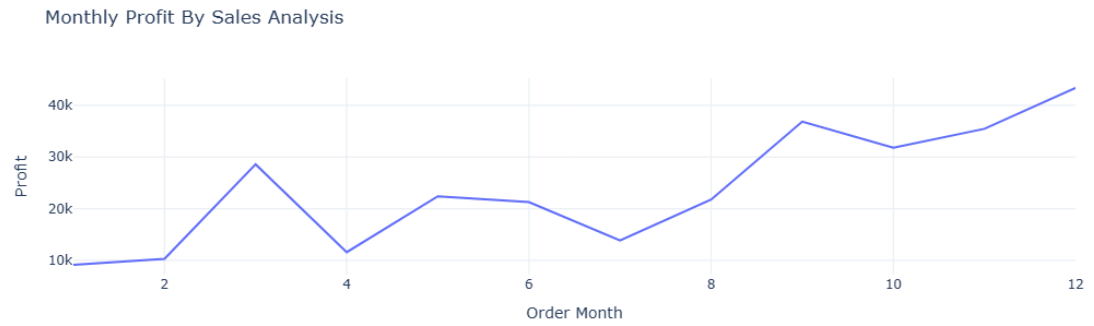
Insight:

Technology and Office Supplies categories contribute the highest sales, indicating strong demand in these segments.

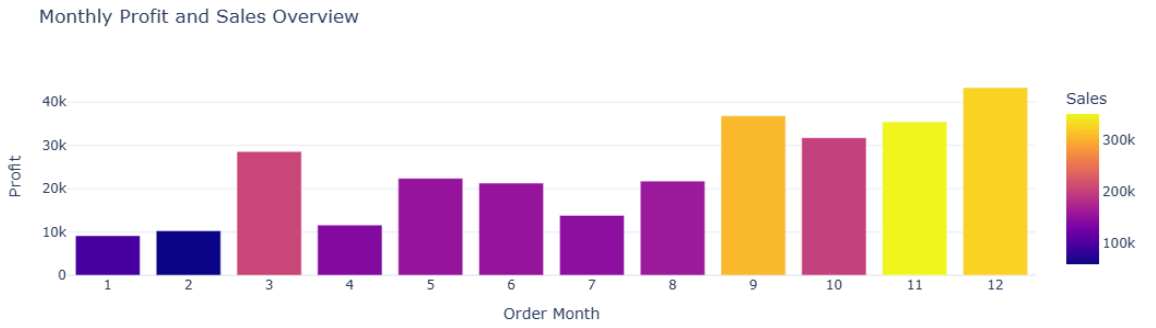
In [29]:
```
sales_by_sub_category= (data.groupby(['Category','Sub-Category'])['Sales'].sum()
fig=px.bar(sales_by_sub_category,
           x= 'Category',
           y= 'Sales',
           color= 'Sub-Category',
           title= 'Sales By Category And Sub-Category')
fig.show()
```



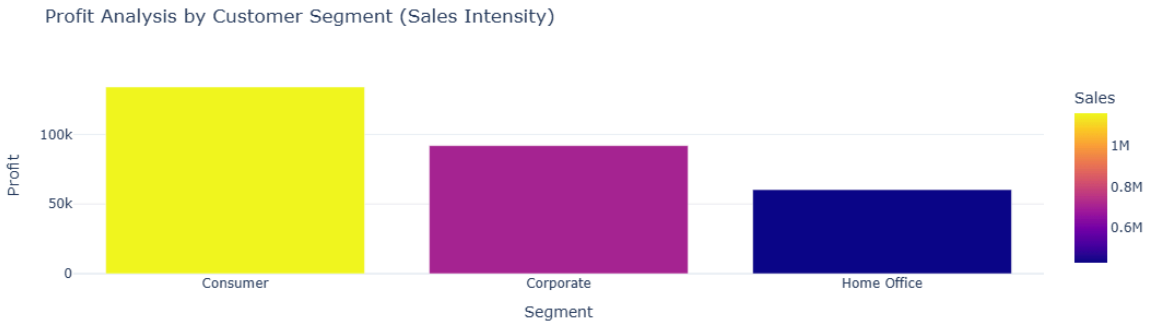
In [31]:
```
monthly_profit_0f_sales= data.groupby('Order Month')['Profit'].sum().reset_index
fig=px.line(monthly_profit_0f_sales,
            x= 'Order Month',
            y= 'Profit',
            title= 'Monthly Profit By Sales Analysis')
fig.show()
```



In [38]:
```
monthly_profit_sales= data.groupby('Order Month')[['Sales','Profit']].sum().rese
fig=px.bar(monthly_profit_sales,
           x= 'Order Month',
           y= 'Profit',
           color= 'Sales',
           title= 'Monthly Profit and Sales Overview')
fig.update_xaxes(
    tickmode='linear',
    tick0= 1,
    dtick= 1
)
fig.show()
```



In [49]:
```
sales_profit_segm= data.groupby('Segment')[['Sales','Profit']].sum().reset_index
fig=px.bar(sales_profit_segm,
           x= 'Segment',
           y= 'Profit',
           color= 'Sales',
           title= 'Profit Analysis by Customer Segment (Sales Intensity)')
fig.show()
```



In [52]:
```
sales_profit_segment = data.groupby('Segment').agg({'Sales': 'sum' , 'Profit': '
sales_profit_segment['Sales_to_profit_Ratio']=sales_profit_segment['Sales']/sale
sales_profit_segment[['Segment','Sales_to_profit_Ratio']]
```

Out[52]:

	Segment	Sales_to_profit_Ratio
0	Consumer	8.659471
1	Corporate	7.677245
2	Home Office	7.125416

📈 Key Insights

- Consumer segment generates the highest profit
- Some sub-categories show negative profit despite high sales
- Profitability varies significantly across months

💡 Business Recommendations

- Focus marketing on high-profit customer segments
- Optimize discounts for loss-making products

- Improve inventory planning based on monthly trends

✅ Conclusion

This analysis provides actionable insights into customer behavior, product performance, and profitability, enabling data-driven decision-making for e-commerce businesses.